CS23336/GE23231/GE23233-IPP/PUP/PSPP

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Status	Finished
Started	Tuesday, 20 October 2026, 6:48 PM
Completed	Tuesday, 20 October 2026, 6:57 PM
Duration	9 mins 16 secs
Grade	**9.00** out of 10.00 (**90**%)

Question **1**
Correct
Mark 1.00 out of 1.00
Flag question

An list contains N numbers and you want to determine whether two of the numbers sum to a given number K. For example, if the input is 8, 4, 1, 6 and K is 10, the answer is yes (4 and 6). A number may be used twice.

Input Format

The first line contains a single integer n , the length of list

The second line contains n space-separated integers, list[i].

The third line contains integer k.

Output Format

Print Yes or No.

Sample Input

7

0 1 2 4 6 5 3

1

Sample Output

Yes

For example:

Input	Result
5 8 9 12 15 3 11	Yes
6 2 9 21 32 43 43 1 4	No

Answer: (penalty regime: 0 %)

```python
n = int(input())
a = list(map(int, input().split()))
k = int(input())

found = False

for i in range(n):
    for j in range(i + 1, n):
        if a[i] + a[j] == k:
            found = True
            break
    if found:
        break

if found:
    print("Yes")
else:
    print("No")
```

	Input	Expected	Got	
✓	5 8 9 12 15 3 11	Yes	Yes	✓
✓	6 2 9 21 32 43 43 1 4	No	No	✓
✓	6 13 42 31 4 8 9 17	Yes	Yes	✓

Passed all tests! ✓

Correct
Marks for this submission: 1.00/1.00.

Question **2**
Correct
Mark 1.00 out of 1.00
Flag question

Two string values S1, S2 are passed as the input. The program must print first N characters present in S1 which are also present in S2.

Input Format:

The first line contains S1.
The second line contains S2.
The third line contains N.

Output Format:

The first line contains the N characters present in S1 which are also present in S2.

Boundary Conditions:

2 <= N <= 10
2 <= Length of S1, S2 <= 1000

Example Input/Output 1:

Input:

abcbde
cdefghbb
3

Output:

bcd

Note:

b occurs twice in common but must be printed only once.

Answer: (penalty regime: 0 %)

```python
s1 = input()
s2 = input()
n = int(input())

ans = ""

for ch in s1:
    if ch in s2 and ch not in ans:
        ans += ch
    if len(ans) == n:
        break

print(ans)
```

	Input	Expected	Got	
✓	abcbde cdefghbb 3	bcd	bcd	✓

Passed all tests! ✓

Correct
Marks for this submission: 1.00/1.00.

Question **3**
Incorrect
Mark 0.00 out of 1.00
Flag question

String should contain only the words are not palindrome.

Sample Input 1

Malayalam is my mother tongue

Sample Output 1

is my mother tongue

Answer: (penalty regime: 0 %)

```python
s = input().split()

ans = []

for word in s:
    w = word.lower()
    if w != w[::-1]:
        ans.append(word)

print(" ".join(ans))
```

	Input	Expected	Got	
✓	Malayalam is my mother tongue	is my mother tongue	is my mother tongue	✓

Your code failed one or more hidden tests.
Your code must pass all tests to earn any marks. Try again.

Incorrect
Marks for this submission: 0.00/1.00.

Question **4**
Correct
Mark 1.00 out of 1.00
Flag question

You are given an `m x n` integer matrix `matrix` with the following two properties:

- Each row is sorted in non-decreasing order.
- The first integer of each row is greater than the last integer of the previous row.

Given an integer `target`, return `True` *if* `target` *is in* `matrix` *or* `False` *otherwise*.

You must write a solution in `O(log(m * n))` time complexity.

Example 1:

1	3	5	7
10	11	16	20
23	30	34	60

```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 3
Output: True
```

Example 2:

1	3	5	7
10	11	16	20
23	30	34	60

```
Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 13
Output: False
```

For example:

Test	Result
`print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))`	False
`print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))`	True

Answer: (penalty regime: 0 %)

Reset answer

```python
def searchMatrix(matrix, target):
    m = len(matrix)
    n = len(matrix[0])

    left = 0
    right = m * n - 1

    while left <= right:
        mid = (left + right) // 2

        row = mid // n
        col = mid % n

        if matrix[row][col] == target:
            return True
        elif matrix[row][col] < target:
            left = mid + 1
        else:
            right = mid - 1

    return False
```

	Test	Expected	Got	
✓	`print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 13))`	False	False	✓
✓	`print(searchMatrix([[1,3,5,7],[10,11,16,20],[23,30,34,60]], 3))`	True	True	✓

Passed all tests! ✓

Correct
Marks for this submission: 1.00/1.00.

Question **5**
Correct
Mark 1.00 out of 1.00
Flag question

Balanced strings are those that have an equal quantity of 'L' and 'R' characters.

Given a balanced string s, split it in the maximum amount of balanced strings.

Return the maximum amount of split balanced strings.

Example 1:

Input:

RLRRLLRLRL

Output:

4

Explanation: s can be split into "RL", "RRLL", "RL", "RL", each substring contains same number of 'L' and 'R'.

Example 2:

Input:

RLLLLRRRLR

Output:

3

Explanation: s can be split into "RL", "LLLRRR", "LR", each substring contains same number of 'L' and 'R'.

Example 3:

Input:

LLLLRRRR

Output:

1

Explanation: s can be split into "LLLLRRRR".

Constraints:

1 <= s.length <= 1000

s[i] is either 'L' or 'R'.

s is a balanced string.

For example:

Test	Result
`print(BalancedStrings('RLRRLLRLRL'))`	4
`print(BalancedStrings('RLLLLRRRLR'))`	3

Answer: (penalty regime: 0 %)

Reset answer

```python
def BalancedStrings(s):
    count = 0
    balance = 0

    for ch in s:
        if ch == 'L':
            balance += 1
        else:
            balance -= 1

        if balance == 0:
            count += 1

    return count
```

	Test	Expected	Got	
✓	`print(BalancedStrings('RLRRLLRLRL'))`	4	4	✓
✓	`print(BalancedStrings('RLLLLRRRLR'))`	3	3	✓

Passed all tests! ✓

Correct
Marks for this submission: 1.00/1.00.

Question **6**
Correct
Mark 1.00 out of 1.00
Flag question

Given an array `nums` containing `n` distinct numbers in the range `[0, n]`, return *the only number in the range that is missing from the array.*

Example 1:

```
Input: nums = [3,0,1]
Output: 2
Explanation: n = 3 since there are 3 numbers, so all numbers are in the range [0,3]. 2 is the missing number i
```

Example 2:

```
Input: nums = [0,1]
Output: 2
Explanation: n = 2 since there are 2 numbers, so all numbers are in the range [0,2]. 2 is the missing number i
```

Example 3:

```
Input: nums = [9,6,4,2,3,5,7,0,1]
Output: 8
Explanation: n = 9 since there are 9 numbers, so all numbers are in the range [0,9]. 8 is the missing number i
```

For example:

Test	Result
`print(missingNumber([3,0,1]))`	2
`print(missingNumber([0,1]))`	2

Answer: (penalty regime: 0 %)

Reset answer

```python
def missingNumber(nums):
    n = len(nums)
    total = n * (n + 1) // 2

    for x in nums:
        total -= x

    return total
```

	Test	Expected	Got	
✓	`print(missingNumber([3,0,1]))`	2	2	✓
✓	`print(missingNumber([0,1]))`	2	2	✓
✓	`print(missingNumber([9,6,4,2,3,5,7,0,1]))`	8	8	✓

Passed all tests! ✓

Correct
Marks for this submission: 1.00/1.00.

Question **7**
Correct
Mark 1.00 out of 1.00
Flag question

Given an array of integers `nums` which is sorted in ascending order, and an integer `target`, write a function to search `target` in `nums`. If `target` exists, then return its index. Otherwise, return `-1`.

You must write an algorithm with `O(log n)` runtime complexity.

Example 1:

```
Input: nums = [-1,0,3,5,9,12], target = 9
Output: 4
Explanation: 9 exists in nums and its index is 4
```

Example 2:

```
Input: nums = [-1,0,3,5,9,12], target = 2
Output: -1
Explanation: 2 does not exist in nums so return -1
```

Constraints:

- $1 <= \texttt{nums.length} <= 10^4$
- $-10^4 < \texttt{nums[i]}, \texttt{target} < 10^4$
- All the integers in `nums` are **unique**.
- `nums` is sorted in ascending order.

For example:

Test	Result
`print(search([-1,0,3,5,9,12],9))`	4

Answer: (penalty regime: 0 %)

Reset answer

```python
def search(nums, target):
    left = 0
    right = len(nums) - 1

    while left <= right:
        mid = (left + right) // 2

        if nums[mid] == target:
            return mid
        elif nums[mid] < target:
            left = mid + 1
        else:
            right = mid - 1

    return -1
```

	Test	Expected	Got	
✓	`print(search([-1,0,3,5,9,12],9))`	4	4	✓
✓	`print(search([-1,0,3,5,9,12],2))`	-1	-1	✓

Passed all tests! ✓

Correct
Marks for this submission: 1.00/1.00.

Question **8**
Correct
Mark 1.00 out of 1.00
Flag question

Write a Python program for binary search.

For example:

Input	Result
1,2,3,5,8 6	False
3,5,9,45,42 42	True

Answer: (penalty regime: 0 %)

```python
a = list(map(int, input().split(',')))
k = int(input())

if k in a:
    print("True")
else:
    print("False")
```

	Input	Expected	Got	
✓	1,2,3,5,8 6	False	False	✓
✓	3,5,9,45,42 42	True	True	✓
✓	52,45,89,43,11 11	True	True	✓

Passed all tests! ✓

Correct
Marks for this submission: 1.00/1.00.

Question **9**
Correct
Mark 1.00 out of 1.00
Flag question

Given two Strings s1 and s2, remove all the characters from s1 which is present in s2.

Constraints

1<= string length <= 200

Sample Input 1

experience
enc

Sample Output 1

xpri

Answer: (penalty regime: 0 %)

```python
s1 = input()
s2 = input()

ans = ""

for ch in s1:
    if ch not in s2:
        ans += ch

print(ans)
```

	Input	Expected	Got	
✓	experience enc	xpri	xpri	✓

Passed all tests! ✓

Correct
Marks for this submission: 1.00/1.00.

Question **10**
Correct
Mark 1.00 out of 1.00
Flag question

Given an list, find peak element in it. A peak element is an element that is greater than its neighbors.

An element a[i] is a peak element if

A[i-1] <= A[i] >=a[i+1] for middle elements. [0<i<n-1]

A[i-1] <= A[i] for last element [i=n-1]

A[i]>=A[i+1] for first element [i=0]

Input Format

The first line contains a single integer n , the length of A .
The second line contains n space-separated integers,A[i].

Output Format

Print peak numbers separated by space.

Sample Input

5

8 9 10 2 6

Sample Output

`10 6`

For example:

Input	Result
4 12 3 6 8	12 8

Answer: (penalty regime: 0 %)

```python
n = int(input())
a = list(map(int, input().split()))

for i in range(n):
    if i == 0:
        if a[i] >= a[i + 1]:
            print(a[i], end=" ")
    elif i == n - 1:
        if a[i] >= a[i - 1]:
            print(a[i], end=" ")
    else:
        if a[i] >= a[i - 1] and a[i] >= a[i + 1]:
            print(a[i], end=" ")
```

	Input	Expected	Got	
✓	7 15 7 10 8 9 4 6	15 10 9 6	15 10 9 6	✓
✓	4	12 8	12 8	✓

	12 3 6 8			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Finish review